NON-STEROIDAL ANTI-INFLAMMATORY DRUGS (NSAIDS) AND PARACETAMOL

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At the end of the lecture, you must be able to:

1. Identify mechanisms of action, the therapeutic uses, adverse effect and contraindication of commonly used analgesics (NSAIDs e.g. aspirin ..and paracetamol).
2. Contrast the actions and toxicity of aspirin, the older non-selective NSAIDs and the COX-2-selective drugs .
3. Describe the actions of paracetamol (acetaminophen), its pharmacokinetic effects, its clinical uses ,adverse effects and toxicity and its management.

Painkiller:

Diffrance between analgesic, antipyretic and antiinflamatory medications

<table>
<tr><th>Antipyretic
Reduce Fever (Central)</th><th>Analgesic
Reduce pain</th><th>Anti-inflamatory
Reduce inflammation</th></tr>
<tr><td colspan="2">Paracetamol (acetaminophen).
Nefopam (Acuapan)</td><td></td></tr>
<tr><td colspan="3">NSAIDs (ibuprofen, diclofenac, Naproxen etc.)
Aspirin (Acetylsalicylic acid).</td></tr>
<tr><td></td><td colspan="2">strong Opioid
Weak Opioid (their anti inflamatory effect is contervesial).</td></tr>
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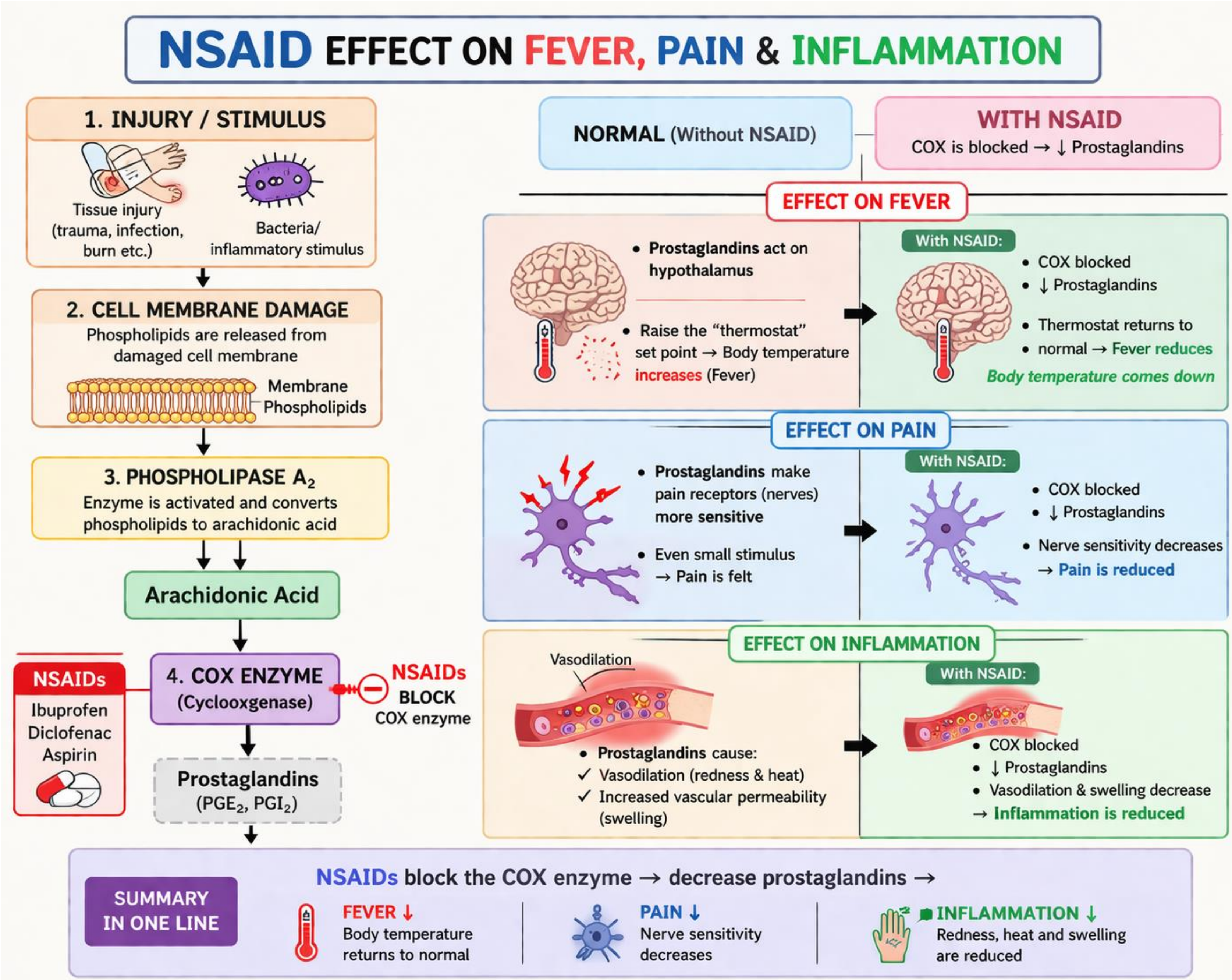

NSAID EFFECT ON FEVER, PAIN & INFLAMMATION
1. INJURY / STIMULUS
Tissue injury (trauma, infection, burn etc.)
Bacteria/ inflammatory stimulus
2. CELL MEMBRANE DAMAGE
Phospholipids are released from damaged cell membrane
Membrane Phospholipids
3. PHOSPHOLIPASE A2
Enzyme is activated and converts phospholipids to arachidonic acid
Arachidonic Acid
NSAIDs
Ibuprofen
Diclofenac
Aspirin
4. COX ENZYME (Cyclooxygenase)
NSAIDs BLOCK COX enzyme
Prostaglandins (PGE2, PGI2)
NORMAL (Without NSAID)
WITH NSAID
COX is blocked → ↓ Prostaglandins
EFFECT ON FEVER
Prostaglandins act on hypothalamus
Raise the "thermostat" set point → Body temperature increases (Fever)
With NSAID:
COX blocked
↓ Prostaglandins
Thermostat returns to normal → Fever reduces
Body temperature comes down
EFFECT ON PAIN
Prostaglandins make pain receptors (nerves) more sensitive
Even small stimulus → Pain is felt
With NSAID:
COX blocked
↓ Prostaglandins
Nerve sensitivity decreases → Pain is reduced
EFFECT ON INFLAMMATION
Vasodilation
Prostaglandins cause:
✓ Vasodilation (redness & heat)
✓ Increased vascular permeability (swelling)
With NSAID:
COX blocked
↓ Prostaglandins
Vasodilation & swelling decrease
→ Inflammation is reduced
SUMMARY IN ONE LINE
NSAIDs block the COX enzyme → decrease prostaglandins →
FEVER ↓
Body temperature returns to normal
PAIN ↓
Nerve sensitivity decreases
INFLAMMATION ↓
Redness, heat and swelling are reduced

Non-steroidal Anti-inflammatory Drugs (NSAIDs)

Analgesic | Antipyretic | Anti-inflammatory

Mechanism of action:

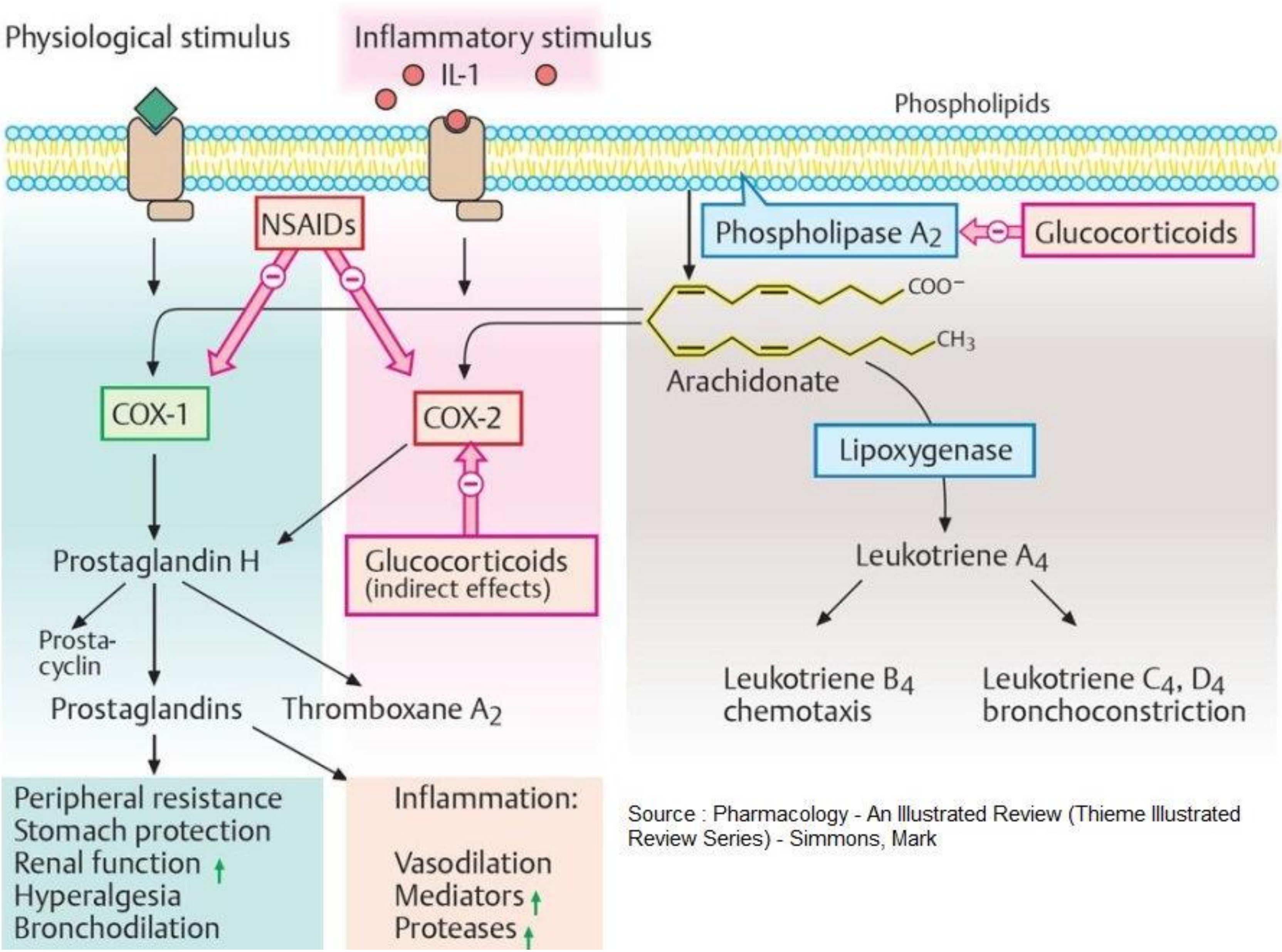
NSAIDs inhibit cyclo-oxygenase enzymes (COX-1 & COX-2) ➔ inhibits the conversion of arachidonic acid to prostaglandins (PGs) + thromboxane A2 (TXA_2).

Results: less PGs ➔ ↓ Pain - ↓ Fever - ↓ Inflammation

COX Isoenzymes

COX-1 is present normally in most tissues (constitutive). It produce ➔ ***protective PGs that is needed for gastric mucosa protection, platelet aggregation and maintain renal blood flow.***

COX-2 is inducible (formed in inflammation) and also constitutive in some tissues like kidney and brain.



Source : Pharmacology - An Illustrated Review (Thieme Illustrated Review Series) - Simmons, Mark

➢ Classification of NSAIDs:

A. Non-selective COX inhibitors (inhibits both COX-1 & COX-2):

1. Acetylsalicylic acid ***(aspirin):***

   the prototype (***irreversible*** inhibition of PGs)

2. Other members e.g., diclofenac, ibuprofen, indomethacin etc (***reversible*** inhibition of PGs).

B. Selective COX-2 inhibitor e.g., celecoxib

I. ACETYLSALICYLIC ACID (ASPIRIN)

EFFECTS:

1. Low Dose (**75-150 or 81 mg/d**): ***Anti-platelet*** (↓ TXA_2 synthesis)

2. Intermediate dose (**300-650mg** every 4–6 hours or **4 g/day**):

a. ***Analgesic***: decrease in PG, leading to central and peripheral effects (dual effect, but the peripheral effect is more significant) mild pain (not effective in angina and cancer pain).

b. ***Antipyretic***: Aspirin reduces fever by ↓ **prostaglandin (PGE_2) production in the hypothalamus**, which **resets the thermoregulatory set point to normal** → activates heat loss (**vasodilation & sweating**).

Aspirin does not affect normal temperature.

3. High dose (**4-8 g/d**): Anti-inflammatory .

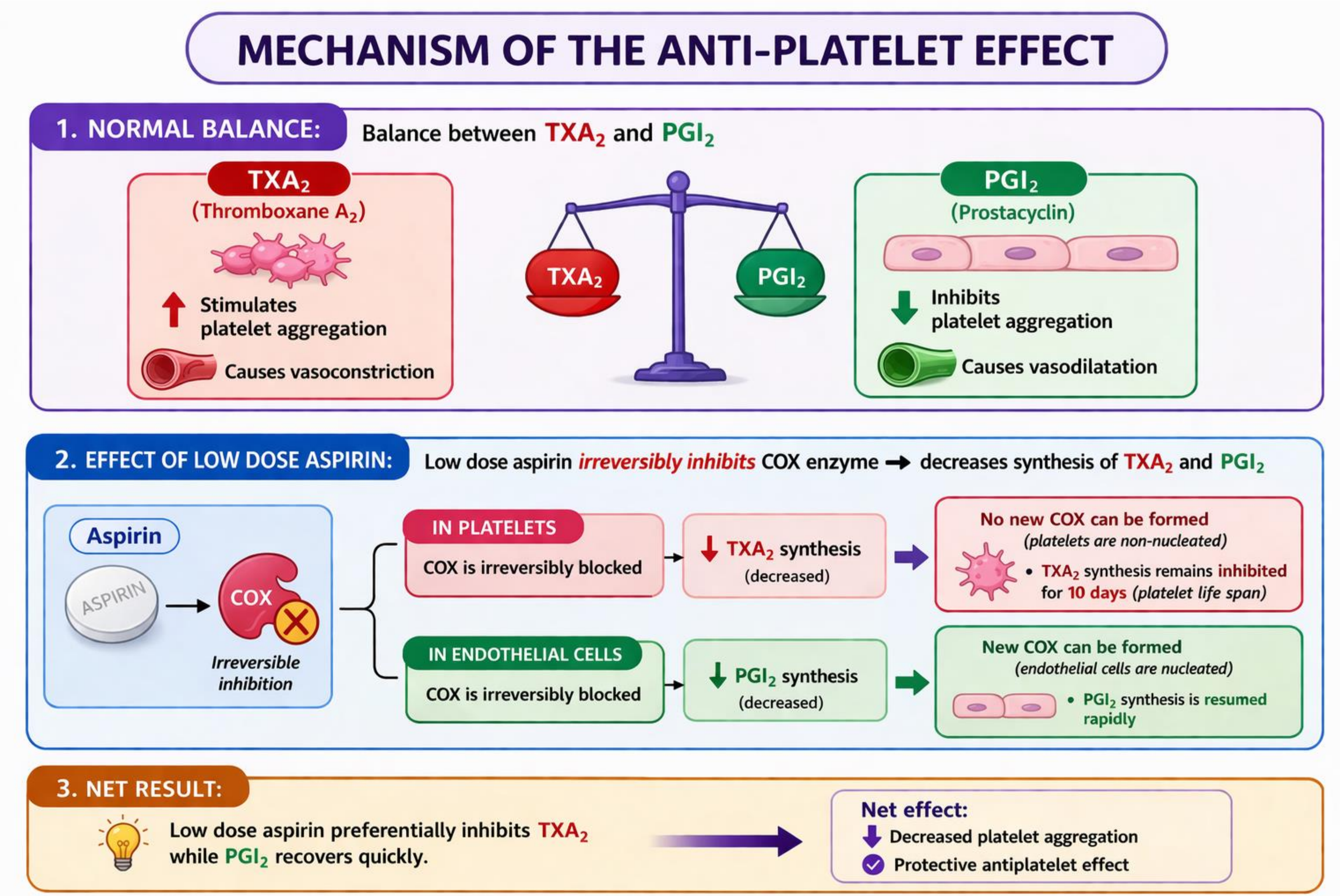
Aspirin is now used mainly for its antiplatelet effect, while safer drugs like Ibuprofen and paracetamol are preferred for pain due to aspirin's higher risk of GI bleeding.

Mechanism of the anti-platelet effect

- **Normally there is a balance between TXA_2 and PGI2: TXA_2 stimulates platelet aggregation and causes vasoconstriction while PGI2 inhibits platelet aggregation and causes vasodilatation.**

  Low dose aspirin irreversibly inhibits COX enzyme → decrease synthesis of ***TXA2 in platelets and PGI2 in the endothelial cells.***

- The endothelial cells can form new COX → PGI2 synthesis is resumed rapidly.
- The platelets are non-nucleated thus they cannot form new COX → TXA2 synthesis is inhibited for 7-10 days (platelet life span).


MECHANISM OF THE ANTI-PLATELET EFFECT
1. NORMAL BALANCE:
Balance between TXA_2 and PGI_2
TXA_2
(Thromboxane A_2)
Stimulates platelet aggregation
Causes vasoconstriction
TXA_2
PGI_2
PGI_2
(Prostacyclin)
Inhibits platelet aggregation
Causes vasodilatation
2. EFFECT OF LOW DOSE ASPIRIN:
Low dose aspirin *irreversibly inhibits* COX enzyme → decreases synthesis of TXA_2 and PGI_2
Aspirin
ASPIRIN
COX
Irreversible inhibition
IN PLATELETS
COX is irreversibly blocked
↓ TXA_2 synthesis (decreased)
No new COX can be formed
(platelets are non-nucleated)
• TXA_2 synthesis remains inhibited for 10 days *(platelet life span)*
IN ENDOTHELIAL CELLS
COX is irreversibly blocked
↓ PGI_2 synthesis (decreased)
New COX can be formed
(endothelial cells are nucleated)
• PGI_2 synthesis is resumed rapidly
3. NET RESULT:
Low dose aspirin preferentially inhibits TXA_2 while PGI_2 recovers quickly.
Net effect:
Decreased platelet aggregation
Protective antiplatelet effect

Q: Why does aspirin have an irreversible antiplatelet effect, unlike other NSAIDs?

- Aspirin **irreversibly inhibits COX** by forming a **covalent (acetyl) bond.**
- Other NSAIDs **bind reversibly** to COX (temporary inhibition- hours)
- Platelets **cannot synthesize new COX** → effect lasts **7–10 days**

➢ **Pharmacokinetics :**

- Rapidly absorbed from stomach and upper small intestine (acidic medium). Rapidly hydrolyzed into acetic acid & salicylate by esterases in blood and tissues.
- Salicylate is *highly bound to plasma proteins* (albumin) ➔ ***displaces*** warfarin (an oral anticoagulant) → increase warfarin serum level→ bleeding. (DD interaction)

Cont.

Elimination of salicylate is dose-dependent:

- At **low doses (< 4 g/day)** → follows first-order kinetics ($t_{1/2} \approx 4$ hours).
- At **high doses (> 4 g/day)** → metabolic pathways become **saturated**, leading to **prolonged half-life (> 15 hours)** and drug accumulation → **toxicity** (fever, confusion, metabolic acidosis).

Clinical note:

- **Alkalinization of urine** increases salicylate ionization → enhances renal excretion → used in **management of toxicity**.

Indications of NSAIDs:

1. **Anti-thrombotic**: **only aspirin** is used for Prophylaxis against thrombosis (low dose).
2. **Mild to moderate pain** e.g., dental pain and headache.
3. **Severe pain** e.g., postoperative & cancer pain (**added to opioids to decrease their dose).**

4- **Antipyretic** by decreasing concentration of prostaglandins in the hypothalamus to overcome the endogenous pyrogenic response (paracetamol is preferred).

5-**Musculoskeletal inflammatory disorders** e.g., arthritis.

6- **Indomethacin** : is indicated for **patent ductus arteriosus closure.**

➢ Adverse Effects of NSAIDs:

A. Adverse Effects Common to all NSAIDs (especially in the elderly):

1. **GIT upset and ulceration** (with ↑ risk of bleeding)
2. **Nephrotoxicity**
3. **Hypersensitivity**: Skin rash, rhinitis, asthma.
4. **Fluid retention,** hypertension, edema.

B. Adverse Effects Specific to Aspirin:

1. **Bleeding:** Due to its antiplatelet effect → aspirin should be **discontinued 1 week before surgery**.
2. **Low doses**: Cause **hyperuricemia** by reducing uric acid excretion.

(Competes with uric acid for secretion) → Gout

3. **High doses**: Produce a **uricosuric effect** → increase uric acid excretion. (Prevents it from going back into blood) → uric acid kidney stones

4. **Reye's syndrome:** A serious condition causing **encephalopathy and liver damage** in children with viral infections → **contraindicated in children <12 years.**
5. **Acute toxicity**: hyperventilation resulting from direct respiratory center stimulation, leading to **respiratory alkalosis then metabolic acidosis.**
6. **Chronic toxicity (salicylism): prolonged large** doses → dizziness, tinnitus, nausea, and vomiting.

➢ **Contraindications of aspirin: (Use Paracetamol instead of aspirin in these cases):**

1. Allergy to aspirin.
2. GIT disorders (peptic ulcer)
3. Renal problems.
4. Gout.
5. Viral infections in children (to avoid Reye's syndrome).
6. Bleeding disorders.
7. Pregnancy (Taking NSAIDs after 19 weeks of pregnancy could cause rare but serious kidney problems in fetuses).


NDC 0574-7034-12
Aspirin
Suppositories, USP
300mg (5 grains)
UNIT DOSE
12
Paddock
Aspirin


QC
QUALITY CHOICE
6 pack
**Compare to ECO
active ingredient
Enteric Coated
Aspirin 325 mg (NSAID)
Regular Strength • Aspirin Regimen
Aspirin 325 mg | Pain Reliever
Relieves: Arthritis Pain & Inflammation
100 Tablets
actual size


Aspirin Tablets
75 mg Enteric Coated
For Oral Use
75
mg
Can help prevent further
heart attacks and strokes
in patients with previous
history of these conditions.
Can also be used following
bypass surgery.
28 Tablets
TEVA


TopCare
SEE NEW WARNINGS INFORMATION
NDC 36800-440-12
ADULT LOW DOSE
Aspirin 81mg
• Pain Reliever (NSAID)*
Supports
Children's
Miracle Network
Hospitals
COMPARE TO ASPIRIN
REGIMEN BAYER® 81mg
active ingredient**
Enteric Safety-Coated
120 TABLETS DELAYED RELEASE ASPIRIN

II. Other non-selective NSAIDS

1. Ibuprofen:

- ***First-choice in inflammatory joint disease***
- Less effective but better tolerated (less side effects) than other NSAIDs.
- Best NSAIDs in children.
- May cause HTN in women >30y (unknown mechanism)

2. Diclofenac 75mg:

- Good efficacy (stronger than ibuprofen)
- Relatively low side effects (more than ibuprofen but less than indomethacin).
- Less gastric irritation than aspirin, but more nephrotoxic (2 ampules/day can decrease renal blood flow)
- Best NSAIDs for dysmenorrhea: Better than ibuprofen and indomethacin.

3. Indomethacin:

- More potent (higher efficacy) but more toxic (neurotoxicity).
- Used for closure of ductus arteriosus or cause early closure of ductus arteriosus (avoid in late pregnancy).
- ***Serious adverse effects*** (aplastic anemia, neutropenia or thrombocytopenia).
- limited use (e.g., in Acute gouty arthritis and Rheumatoid arthritis).

4. Oxicam:

- long half life 36hr.
- Used in chronic pain

III. Selective COX-2 inhibitors (e.g., Celecoxib):

- They **do not inhibit COX-1 enzyme.**
- Thus, they have **less adverse effects on the GIT (less risk of gastric ulcer).**
- They have the same anti-inflammatory activity as non-selective members
- **Used in:** musculoskeletal inflammatory disorders (e.g. osteoarthritis, rheumatoid arthritis)
- **Adverse Effects**

1. Nephrotoxicity.
2. Skin rash with celecoxib.
3. **Increase incidence of myocardial infarction.**

Although **selective COX-2 inhibitors** are more specific and cause **less gastric irritation**, they are **not preferred over conventional NSAIDs** because they **increase cardiovascular risk**.

WHY SELECTIVE COX-2 INHIBITORS ARE NOT BETTER THAN NSAIDs

NSAIDs (Non-selective)

Block **BOTH** COX-1 + COX-2

COX-1 (constitutive) + COX-2 (inducible)

Effect
- ↓ Pain, inflammation, fever
- ↓ Platelet aggregation (TXA_2)
- ↓ Gastric protection (PGE_2)

GI Risk
Higher risk of ulcer & bleeding

Thrombosis Risk
Neutral or slightly anti-thrombotic (TXA_2 decreased)

Selective COX-2 Inhibitors (e.g. celecoxib)

Block COX-2 **ONLY**

COX-1 (spared) + COX-2 (blocked)

Effect
- ↓ Pain, inflammation, fever
- Platelet function unchanged
- Gastric protection maintained

GI Risk
Lower risk of ulcer & bleeding

Thrombosis Risk
Increased risk of thrombosis

WHY THROMBOSIS RISK INCREASES WITH COX-2 SELECTIVE?

Normal Balance

COX-1 in platelets	COX-2 in endothelium
↓	↓
TXA_2	PGI_2 (prostacyclin)
Pro-thrombotic	Anti-thrombotic

With COX-2 Selective Inhibitor

COX-1 in platelets	COX-2 blocked
↓	↓
TXA_2 (still produced)	↓ PGI_2 (reduced)

Result:

Unopposed TXA_2 → ↑ Platelet aggregation → ↑ Thrombosis risk (MI, Stroke)

Feature	NSAIDs	Selective COX-2 Inhibitors
COX Enzymes Blocked	COX-1 + COX-2	COX-2 only
GI Safety	Higher GI toxicity	Lower GI toxicity
Platelet Effect	Inhibits platelets (↓ TXA_2)	No inhibition of platelets
Thrombosis Risk	Neutral / Slightly protective	Increased risk

Bottom line: Selective COX-2 inhibitors are NOT better overall than NSAIDs because their better GI safety is offset by a higher risk of cardiovascular thrombosis.

Paracetamol (acetaminophen)

Mechanism of Action:

Paracetamol works mainly in the **brain (central action)** by **inhibiting prostaglandin synthesis (COX)** → reduces **pain and fever**.
Its exact mechanism is not fully understood but may also involve **serotonin pathways** and **endocannabinoids**.

Bioavailability	**70-90%**	**Peak**	**Oral 30min to 1h / IV 8-10min**
Protein binding	Negligable	Duration	2-3 h
distribuation	rapidly and evenly throughout most tissues	metabolism	Liver
excreation	Mostly renal, some biliary		
interaction	Isoniazid increase the incidence of hepatotoxicity as it increase the accumulation of NAPQI by 70% by inhibiting the formation of glutathione. Paracetamol increases INR during warfarin therapy (max allowed dose 2 g /week). Alcoholism and liver disease may increase drug-induced hepatotoxicity.		

PARACETAMOL (ACETAMINOPHEN)

NOT NSAID

Effects:

It has analgesic, antipyretic but ***weak anti-inflammatory effects.***

Uses:

- Widely used analgesic and antipyretic drug especially in case of contraindications to aspirin.
- First-line for fever/pain in:
    - Children (no Reye's syndrome risk).
    - Pregnant women (safe in all trimesters).
    - Patients with aspirin contraindications (e.g., peptic ulcers, bleeding disorders, viral infections).

Metabolism and toxicity of paracetamol :

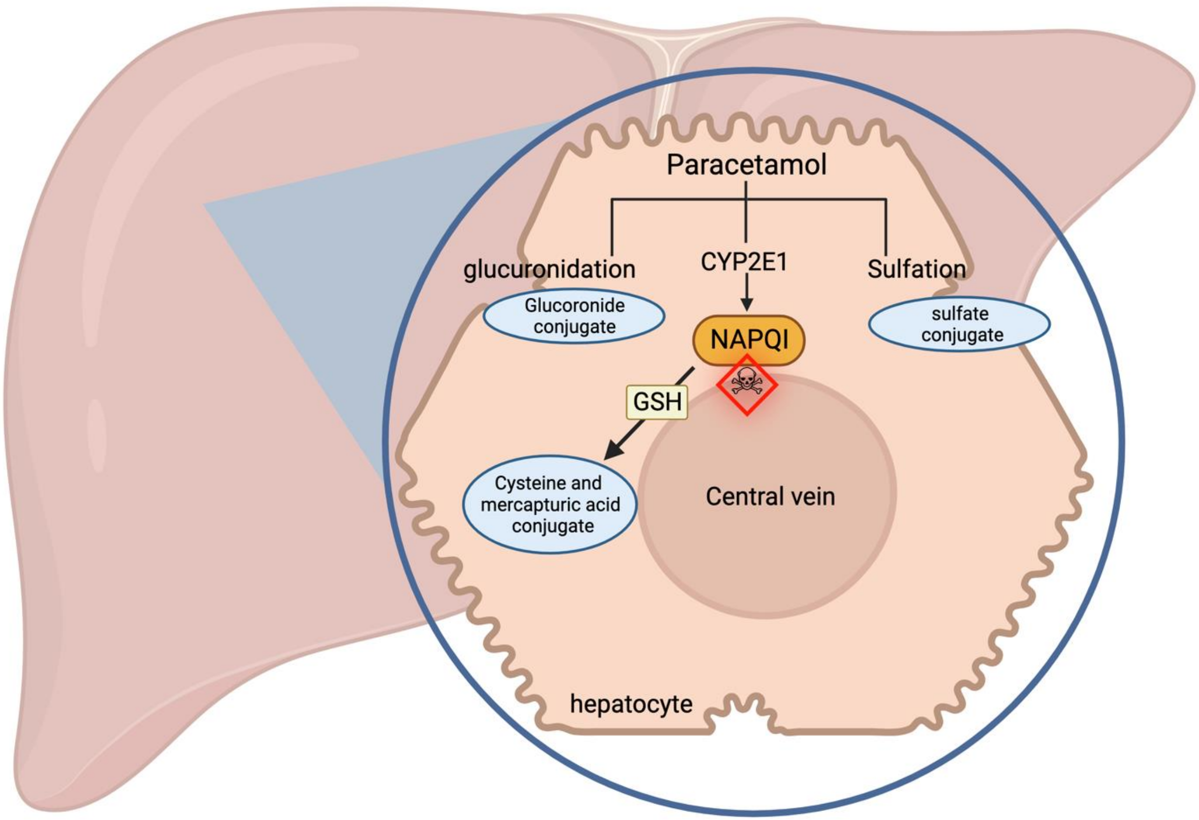
- In therapeutic doses, paracetamol is metabolized in liver by ***two steps***:

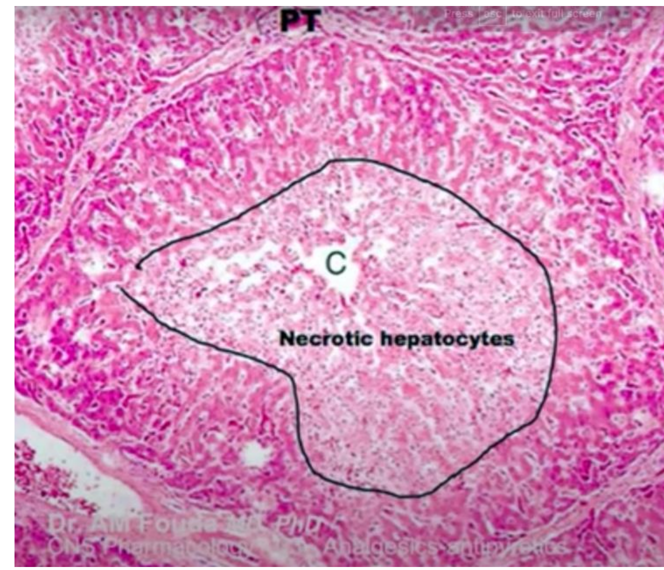
1) 95% is converted by the conjugating enzymes to inactive metabolites.

2) 5% is converted by CYP 450 to an active hepatotoxic metabolite which is then conjugated with glutathione to form inactivate metabolite.

- High doses → formation of toxic metabolite (**NAPQI**) → **liver damage**

NAPQI (ACETYL-*P*-BENZOQUINONE IMINE)





Paracetamol liver toxicity tretment protocol

G	Gastric lavage.
S	Sulfahydryl group drug (-SH) (acetylcysteine) to increase GSH.
H	Hemodialysis.

Administration of glutathione (GSH) itself is ***not effective*** because it cannot cross cell membranes.

An 8-year-old girl has a fever and muscle aches from a presumptive viral infection. Which one of the following drugs would be most appropriate to treat her symptoms?

A. Acetaminophen.

B. Aspirin.

C. Celecoxib.

D. Codeine.

E. Indomethacin.

Which of the following is an analgesic and antipyretic drug that lacks an anti-inflammatory action?

(A) Acetaminophen

(B) Celecoxib

(C) Colchicine

(D) Indomethacin